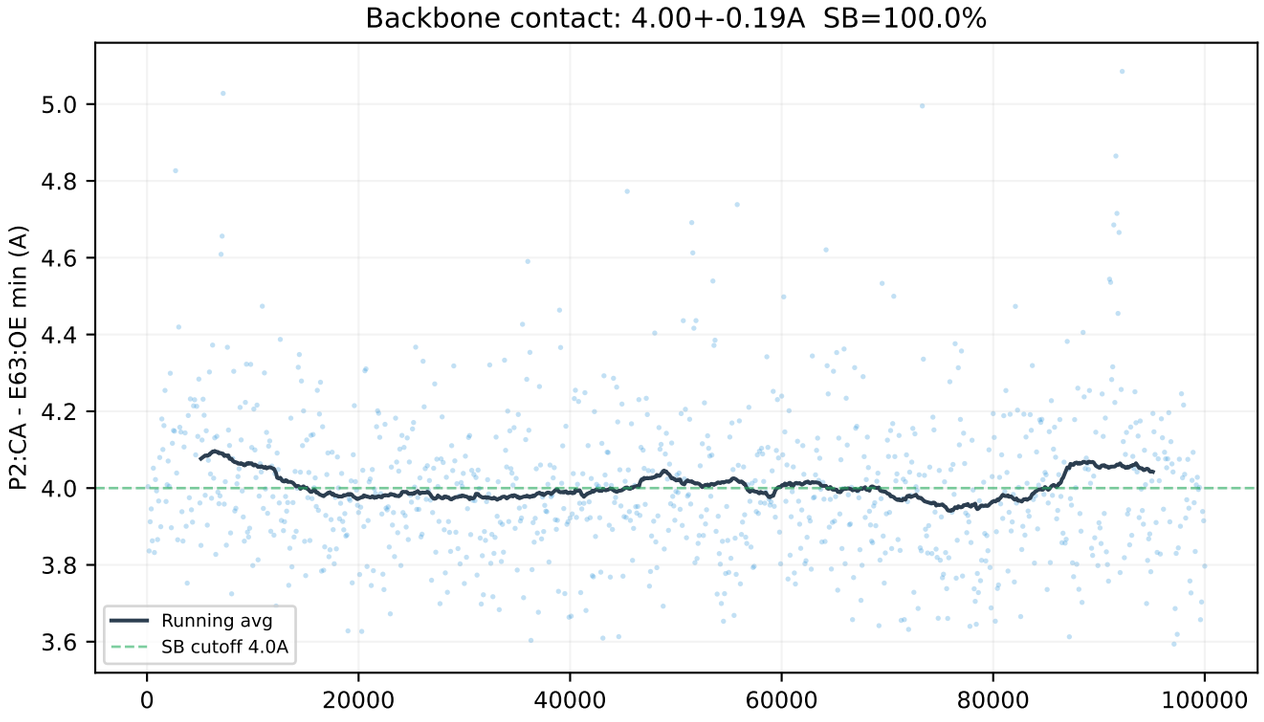
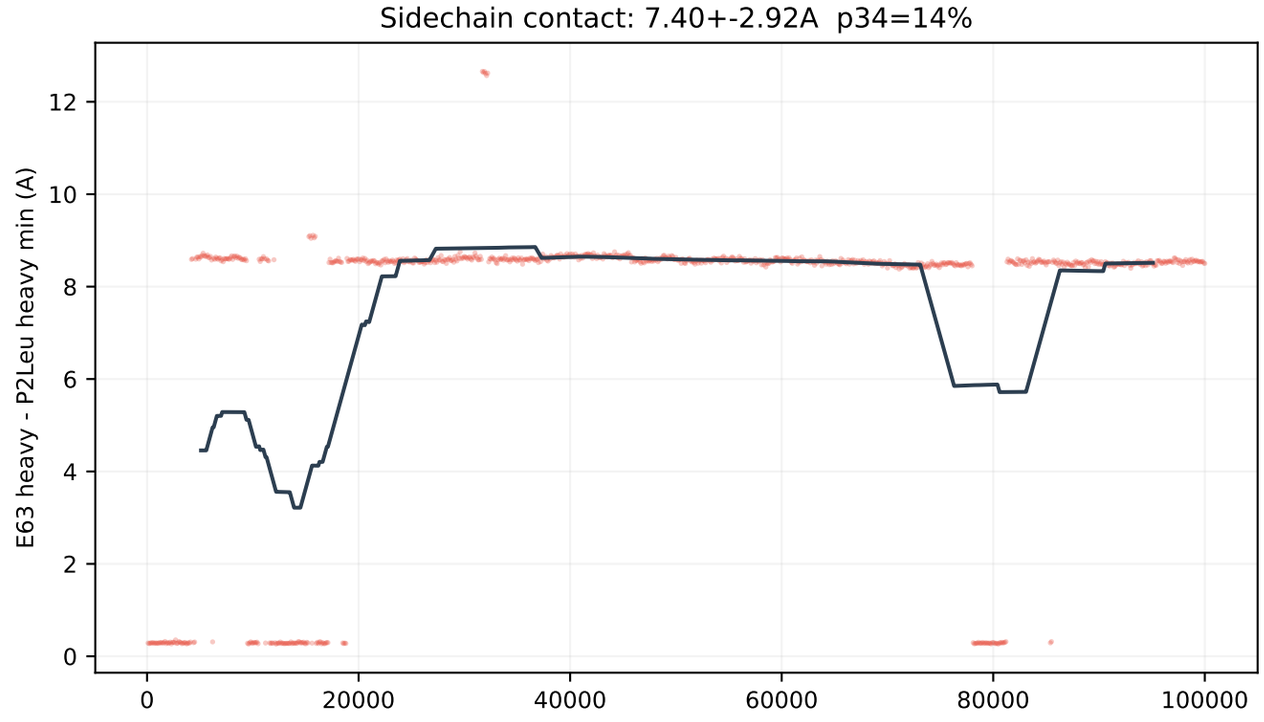
1DUZ K-E63 Salt Bridge — Interface Interaction Analysis (100ns)






```
E63-P2Leu Interaction Mode
==============================

Primary contact (backbone):
  P2:CA - E63:OE  = 4.00+-0.19A
  Salt bridge:     100.0% occupancy

Secondary contact (sidechain):
  E63 heavy - P2Leu heavy = 7.4+-2.9A
  (<3.5A: 14%)

Interpretation:
- E63 carboxyl interacts with P2
  peptide backbone (CA/OE contact~4A)
- P2 leucine sidechain projects into
  MHC B-pocket (not contacting E63)
- This is typical for MHC-I: conserved
  E63 anchors peptide backbone at P2
```

```
Interface H-bonds (A-C): 22 unique

Stability Summary:
  CA-E63: STABLE
  Temp:   205+-1K
  Energy: -292277+-243

MM-GBSA: -19.6 kcal/mol
RMSF:    MHC=1.73A  b2m=3.6A  peptide=4.48A

Replicas running: R2(seed=42), R3(seed=99)
```